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ShieldsUp and HaveIBeenPwned Assignment
Cyber Risk Management
CIDM 6341

Question 1: what did you do?

Provide a description of what you did. In particular, I want to know what you did to research the subject of the report. In most cases the value of the research is in the journey more than the result, same here, the better/clearer/insightful your write up is the better it scores.

In this assignment, I was tasked with running a vulnerability assessment of my home network and personal email address. I followed the provided instructions and went to the ShieldsUp website, where I ran the common ports probe. Once this was complete, I printed the results into a PDF and repeated the process for the service port scan. I then went through my files to find where I had done the ShieldsUp ports scans in the course CIDM6340 Network Management and Information Security (CIDM6340) which I had taken in a prior semester. I did this to review my previous results, so that I could compare the results I had gotten recently to what I had been assessed previously to see how my network security has changed, if at all, since the last time I ran the common ports probe and the service ports scan. In my analysis of the results, I determined there are several ports that were showing as scanned differently. To determine which ports they are, what these ports do, and to assist with researching why these ports would have changed, I used the ShieldsUp website. I was able to select the squares, representing these ports, which took me to another page that provided the port number, as well as background and additional information about that port. This was a good starting point for gaining insights on each of these ports, from which I was able to go do additional research as it was needed. I did this research by referring to class PowerPoints and lectures; however, this was not the favoured method as it was more time consuming to find the exact time in a lecture video, or place in the PowerPoint that I was looking for. Another, and preferable method used, was to ask Google's AI, Gemini. I started with asking Gemini to get a basic answer and then turned to performing my own research through internet searches. I found reliable websites which provided more detail on these topics, as well as further perspective and understanding on the topics. I also verified the sources that Gemini used, to ensure that the answers being provided were from websites that were reliable. I continued this process throughout the ShieldsUp analysis and as I wrote about my results.

For the second part of this assignment, I started out by going to the Have I Been Pwned? Website as directed. Once there, I entered my personal email address – `jaaldakkour@gmail.com` - and my school email address – `jaaldakkour1@buffs.wtamu.edu` - to the search bar one at a time. I then took a screenshot of the results and added them to this report for analysis. I also used the haveibeenpwned website to see if my password was documented anywhere. I checked several passwords.

Question 2: what were the results?

Summarize and explain what your findings mean. Provide any printouts of products produced, this could be a report, a map, a table, etc.

The results of the ShieldsUp scan came back as “failed” (once again), which means that there were ports that responded to the probe from the ShieldsUp scan as either “Open” or “Closed.” What was learned from doing this scan previously, is that when a port responds to a scan as Open, it is stating that it is willing to receive connections and/or data packets. This is a major security risk to a network, as this shows hackers a very easy way to infiltrate a network, and exactly where to start. A Closed response to probe, while better than being entirely open, is still a security risk. A closed response means that a port is not receiving connections and/or data packets, and that it is rejecting attempts at connecting. This is good in that it means that the port is blocking everyone from accessing and connecting to the port; however, in terms of security, it is still not good as it responds to the probe and notifies the user sending out the probe that there is a port there, and as such a device. Ideally, this would not happen, and rather the only results would be that all ports are in “Stealth” mode, as this would mean that a hacker searching for a device would not find one. Stealth mode occurs when there is no response to the probe and is the result of when a firewall is in place. This is the ideal response to an outside network probing. The results from the current and previous common port scans are shown below:

Current Common Port Scan Results:

http://www.grc.com/sec/847b1dd42e

FAILED

**TruStealth
Analysis**

FAILED

Selected TCP Ports: **FAILED (25/26)** -- As detailed in this report below, one or more of your system's ports actively responded to our deliberate attempts to establish a connection. It is generally possible to increase your system's security by hiding it from the portfolio of potentially hostile hackers. Please see the details presented by the specific port data below, as well as the various resources on the left, and in our extensive helpdesk and other associated resources.

Unolicited Port: **FAILED** -- No Internet packets of any port were received from your system as a side effect of our attempts to elicit some responses from any of the ports listed above. Doing so, while some security experts believe that such is worth attempting to "loot" prior to the port, that resulting information, but your system returned none above. (Except for the fact that not all of the ports are completely invisible to these tests.)

Ping Reply: **RECEIVED (25/26)** -- Our system REPLIED to our Ping (ICMP Echo) requests, resulting in replies on the Internet. Most servers should not be configured to block, deny, and reject such ping requests in order to better hide systems from hackers. This is a highly recommended action "You" it being the object of your system's efforts, and to ensure systems prior to further exploration.

Port	Service	Status	Security Implications
8	ircd	Closed	Your computer has responded that this port exists but is currently closed to connections.
21	FTP	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
23	SSH	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
25	Telnet	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
26	SMTP	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
79	Finger	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
80	HTTP	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
135	RPC2	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
133	IDENT	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
110	POP3	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
135	RPC	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
139	Net BIOS	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!

http://www.grc.com/sec/847b1dd42e

1/2

GRC (ShieldsUP) -- Common Ports Probe			
10/4/25, 9:47 AM			
143	IMAP	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
389	LDAP	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
443	HTTPS	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
445	MSFT DC	Stealth	There is NO EVIDENCE WHATSOEVER that a port (or even any computer) exists at this IP address!
1002	irc-66	Closed	Your computer has responded that this port exists but is currently closed to connections.
1024	DCOM	Closed	Your computer has responded that this port exists but is currently closed to connections.
1025	IMAP	Closed	Your computer has responded that this port exists but is currently closed to connections.
1026	IMAP	Closed	Your computer has responded that this port exists but is currently closed to connections.
1027	IMAP	Closed	Your computer has responded that this port exists but is currently closed to connections.
1028	IMAP	Closed	Your computer has responded that this port exists but is currently closed to connections.
1029	IMAP	Closed	Your computer has responded that this port exists but is currently closed to connections.
1030	IMAP	Closed	Your computer has responded that this port exists but is currently closed to connections.
1720	RTSP	Closed	Your computer has responded that this port exists but is currently closed to connections.
5000	UPnP	Closed	Your computer has responded that this port exists but is currently closed to connections.

(Gibson Research Corporation)

Common Port Scan from CIDM6340:

This textual summary may be printed, or marked and copied for subsequent pasting into any other application:

```
-----
GRC Port Authority Report created on UTC: 2024-08-17 at 17:52:31

Results from scan of ports: 0, 21-23, 25, 79, 80, 110, 113,
                           119, 135, 139, 143, 389, 443, 445,
                           1002, 1024-1030, 1720, 5000

  0 Ports Open
 12 Ports Closed
 14 Ports Stealth
-----
26 Ports Tested

NO PORTS were found to be OPEN.

Ports found to be CLOSED were: 0, 21, 1002, 1024, 1025, 1026,
                               1027, 1028, 1029, 1030, 1720,
                               5000

Other than what is listed above, all ports are STEALTH.

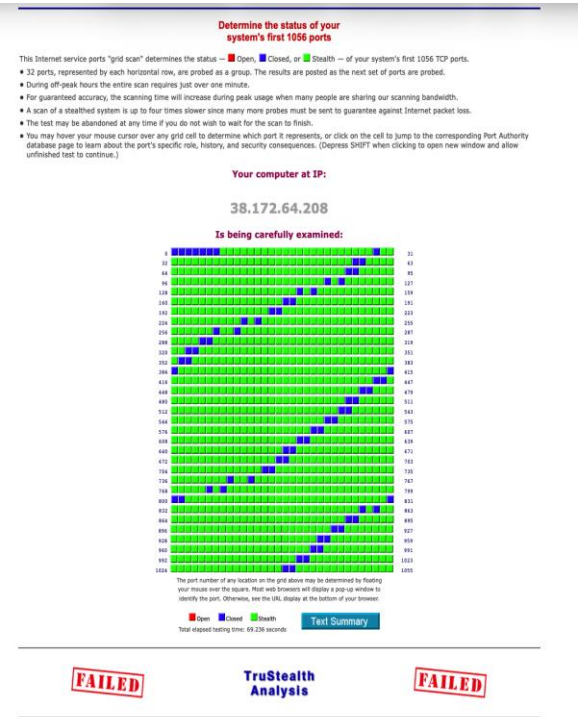
TruStealth: FAILED - NOT all tested ports were STEALTH,
                  - NO unsolicited packets were received,
                  - A PING REPLY (ICMP Echo) WAS RECEIVED.
-----
```

(Aldakkour, 2024)

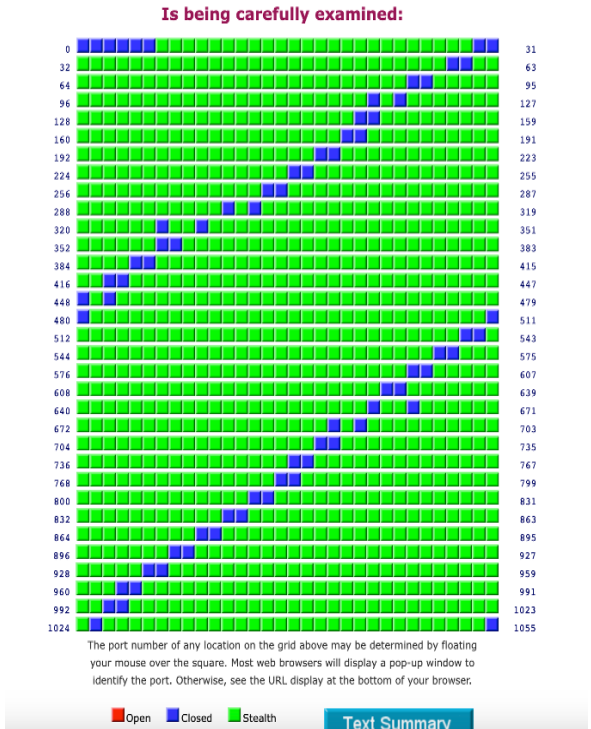
The results from the common port scan differed from when it was done before to now. In both scans, no ports were shown open. The difference comes in with the number of ports that showed up as closed and in stealth. In the current scan, many of the ports which were previous showing as closed still are, however port 21 – which is the port for file transport protocol (FTP) – is showing as in stealth mode now whereas before it was showing up as “closed.” In my research, I investigated reasons why this port and not others would have changed into stealth mode.

The following results are from the service port scan that was performed using ShieldsUp. This scan was also performed in CIDM6340. Both results from CIDM6340 and from the current scan are shown below to continue to be compared.

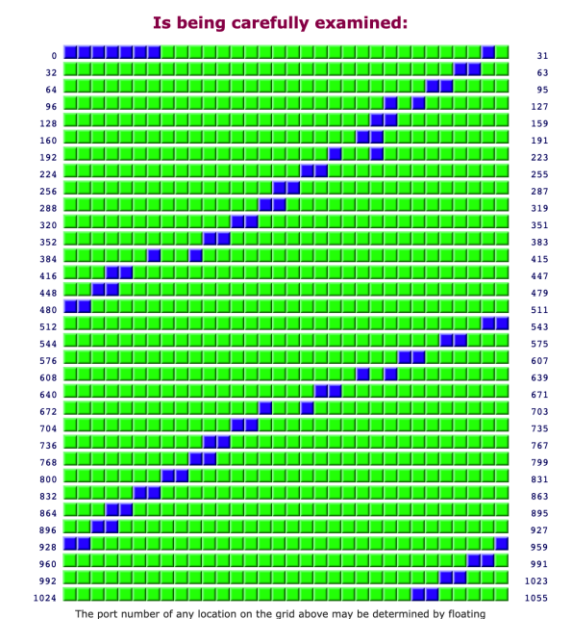
CIDM6355 Service Port Scan Results
(Gibson Research Corporation):



CIDM6340 Service Port Scan Results
(Aldakkour, 2024):



CIDM6355 Service Port Scan Results (Second Recent Run)
(Gibson Research Corporation):



The results show that there are several ports that yielded different results. The only ports that are shown to have held the same results are ports 0 through 5. All other ports had differing results from the original ShieldsUp testing and analysis. When the ShieldsUp test was run again recently, the test yielded different results from both the test from CIDM6340 and from the test run previously for this course. I have done research on reasons why that is which is included in the next section.

Have I Been Pwned Results:

Personal Email:



Email Breach History

Timeline of data breaches affecting your email address

7

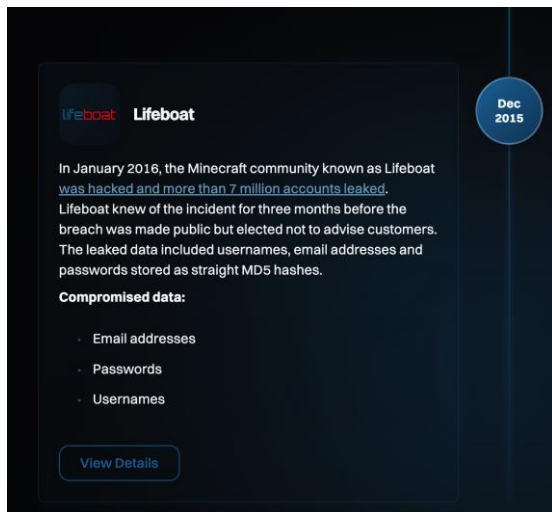
Data Breaches

Oh no — pwned! This email address has been found in multiple data breaches. Review the details below to see where your data was exposed.

 **Stay Protected**

Get notified when your email appears in future data breaches

[Notify Me](#)



 **Lifeboat**

Dec 2015

In January 2016, the Minecraft community known as Lifeboat was hacked and more than 7 million accounts leaked. Lifeboat knew of the incident for three months before the breach was made public but elected not to advise customers. The leaked data included usernames, email addresses and passwords stored as straight MD5 hashes.

Compromised data:

- Email addresses
- Passwords
- Usernames

[View Details](#)

Sep
2016

LEET Leet

In August 2016, the service for creating and running Pocket Minecraft edition servers known as [Leet](#) was reported as having suffered a data breach that impacted 6 million subscribers. The incident reported by Softpedia had allegedly taken place earlier in the year, although the data set sent to HiBP was dated as recently as early September but contained only 2 million subscribers. The data included usernames, email and IP addresses and SHA512 hashes. A further 3 million accounts were obtained and added to HiBP several days after the initial data was loaded bringing the total to over 5 million.

Compromised data:

- Email addresses
- IP addresses
- Passwords
- Usernames
- Website activity

[View Details](#)



Reincubate

In October 2020, the app data company [Reincubate](#) suffered a data breach which exposed a backup from November 2017 (the newest record in the data appeared several months earlier). The data included over 616k unique email addresses, names and passwords stored as PBKDF2 hashes.

Compromised data:

- Email addresses
- Names
- Passwords

[View Details](#)

May
2017

Jan
2018



MyFitnessPal

In February 2018, the diet and exercise service [MyFitnessPal](#) suffered a data breach. The incident exposed 144 million unique email addresses alongside usernames, IP addresses and passwords stored as SHA-1 and bcrypt hashes (the former for earlier accounts, the latter for newer accounts). In 2019, the data appeared listed for sale on a dark web marketplace (along with several other large breaches) and subsequently began circulating more broadly.

Compromised data:

- Email addresses
- IP addresses
- Passwords
- Usernames

[View Details](#)

 Houzz

May 2018

In mid-2018, the housing design website [Houzz](#) suffered a [data breach](#). The company learned of the incident later that year then disclosed it to impacted members in February 2019. Almost 49 million unique email addresses were in the breach alongside names, IP addresses, geographic locations and either salted hashes of passwords or links to social media profiles used to authenticate to the service. The data was provided to HIBP by [dehashed.com](#).

Compromised data:

- Email addresses
- Geographic locations
- IP addresses
- Names
- Passwords
- Social media profiles
- Usernames

View Details

Jan 2019

 **Collection #1**

In January 2019, a large collection of credential stuffing lists (combinations of email addresses and passwords used to hijack accounts on other services) was discovered being distributed on a popular hacking forum. The data contained almost 2.7 *billion* records including 773 million unique email addresses alongside passwords those addresses had used on other breached services. Full details on the incident and how to search the breached passwords are provided in the blog post [The 773 Million Record "Collection #1" Data Breach](#).

Compromised data:

- Email addresses
- Passwords

View Details

 Hot Topic

Oct 2024

In October 2024, [retailer Hot Topic](#) suffered a [data breach that exposed 57 million unique email addresses](#). The impacted data also included physical addresses, phone numbers, purchases, genders, dates of birth and partial credit data containing card type, expiry and last 4 digits.

Compromised data:

- Dates of birth
- Email addresses
- Genders
- Names
- Partial credit card data
- Phone numbers
- Physical addresses
- Purchases
- Salutations

View Details

School Email:

The image shows two screenshots of the Have I Been Pwned website. The top screenshot is the 'Email Breach History' page, which displays '0 Data Breaches' in green. Below this, a message states: 'Good news — no pwnage found! This email address wasn't found in any of the data breaches loaded into Have I Been Pwned. That's great news!'. At the bottom of this section, there is a 'Stay Protected' notification with a bell icon and a 'Notify Me' button. The bottom screenshot is the 'Pwned Passwords' page. It features a search bar with a password field (represented by dots) and a 'Check' button. Below the search bar, a green box with a checkmark icon displays the message: 'Good news — no pwnage found!'. A smaller text below this message says: 'This password wasn't found in any of the Pwned Passwords loaded into Have I Been Pwned. That doesn't necessarily mean it's a good password, merely that it's not indexed on this site.' The website's navigation bar at the top of the bottom screenshot includes links for 'Who's Been Pwned', 'Passwords', 'Notify Me', 'API', 'Demos', 'Pricing', 'About', and a 'Dashboard' button.

The haveibeenpwned results were used to test both my personal email (jaldakkour@gmail.com) and my school email (jaaldakkour1@buffs.wtamu.edu), as well as a couple of different passwords. My school email yielded a result of 0 data breaches. This is good news, in that this email is not included in any data breaches that have been reported and loaded into the haveibeenpwned website. This website was created to collect and securely store “vast amounts of compromised data” to make it so that people can search to see if their emails and/or passwords have been compromised (see reference [7]). How it works is that haveibeenpwned collects data from “public data dumps,” where their team searches the dark web, hacker forums, and other places known for being areas where leaked data can be found, and “official submissions”, from major companies, the FBI, etc (see reference [7]). They then analyse and “vet” the data to ensure it is accurate and truly been leaked before adding it to their collection of leaked information (see reference [7]). This means that, while

ihavebeenpwned is a good website to check if your data has been included in a known data breach, it is not going to be able to inform the user of unknown data breaches. So, there is a possibility that the user's information may still have been leaked, just that the data breach where that has happened has not been discovered or made public information yet.

For my personal email, it was discovered to be a part of seven different data breaches over the span of the past ten years (from 2015 to now). These breaches came from a variety of sources, which include Hot Topic, a retail store, MyFitnessPal, a health app, Reincubate, Houzz, a home design software, and several servers related to Minecraft including Leet and LifeBoat (Hunt, T). Along with providing the data breaches my email was involved in, haveibeenpwned also provides background information about the data breaches, how the haveibeenpwned team discovered the data breaches, and the types of data that was compromised. The passwords that I checked through haveibeenpwned all had positive results. Not one of them was found to have been pwned, which means that the passwords were not found and stored on the haveibeenpwned site.

Question 3: what did you learn?

I can't tell you what you learned. What I will say is that I reward insight. Insights are aha moments (a term in use long before Oprah wanted to copyright it). If you see new ways of doing things, new insights to your thought processes, potential future applications be they personal or work related, crossovers to other topics, these are what I reward more than just telling me you learned lots. I expect you to learn lots, but it isn't till you explain where and what that I see that you really did. Ok, so sometimes you don't learn much. I'll still grade this area high if you tell me why, what you know, how this works on what you've done in the past, etc. Sometimes when you start doing this you see that what you've learned is reinforcing what you've done and sometimes you even have small aha moments. Bottom line is to be reflective, think a few minutes or overnight about what you've done and how it fits into your nomological net (your personal set of knowledge base structure, those theories and beliefs that guide how you evaluate and use knowledge). Then write the section, when I see this done, I always score the section higher.

I have done a similar assignment to this in a prior class, which provided me a base understanding of the topics discussed to start my analysis on. I expanded on my base knowledge in many ways and developed a deeper understanding of these topics. One thing I learned from this was more about stealth ports, open ports, and firewalls. I remembered from doing this assignment previously that a network relies on ports being open, as that is how communication occurs between devices. What was confusing to me and led to me doing further research, was that if a port needs to be open for communication, then how do you keep it secure? Can a port be in two different states at once – such as being open for trusted devices but closed or in stealth mode for devices outside of the network? While my initial follow-up to these questions was that firewalls are involved as the security measure protecting the network, I could not remember exactly how they performed these functions, or if I had done a deep dive into it previously. So, to get a better understanding of how firewalls work with ports, and what exactly occurs, I turned to Google Gemini and the internet to learn more about these things. I first asked Gemini my question regarding port state, and whether it was possible for a port to be in two states at once. The response stated that it technically is not possible, however, that a port can show as being “in different states to different external devices at the same time” (see reference [4]). It discussed “Host’s View” or “Intrinsic State”, which is the actual state in which a port is in regardless of what device is attempting a connecting, and “External View” or “Apparent State” which is when a firewall manipulates the state of a port “depending on the source of traffic” in ways that changes its appearance to outside devices (see reference [4]). To gain further insight and to reassure these ideas are

accurate, I turned to an internet search, where I found a website for the cybersecurity company, Fortinet, that explained this as well. Fortinet's website expanded on how a firewall manipulates the appearance of a port state to devices outside of a network.

Another question that came up because of my comparing the port scan done for CIDM6340 to the current port scan performed in this course is why Port 21, which was previously showing up as "closed," is now showing up in "stealth" mode. I was curious as to why this was, and what changed so that this specific port would now be operating under stealth mode. I followed my previous process of first asking Google Gemini why this is. The response was that this change is the likely result of an update of some kind, whether it is a "router or modem firmware update," a "operating system or software firewall" update, a "change in Internet Service Provider (ISP) hardware or policy," or the introduction of a "new or reconfigured firewall/networking gear" (see reference [5]). I followed this up by performing my own research outside of Gemini, to which the answers I found were the same as those provided by Gemini.

Furthermore, when the service port scan was run through ShieldsUp, there were differences between the scan from CIDM6340 and the current runs, but there were also differences I noted between the two times I ran the scan within a week of each other. I looked at reasons that the port scan would yield different results for different scans in various periods of time. I wondered if it was within my control, that I could directly change to get have these ports consistently respond in stealth mode instead of switching between stealth and closed in different scans. In my research, I found that there are several reasons that this could happen which include the router firewall settings adjusting for perceived threats and rate limiting occurring [8]. This led me down the rabbit hole of researching and learning more about rate limiting. I found the solo.io website the best in terms of summarizing and explaining the topic of rate limiting and why it is important. Rate limiting is a technique used by many items to limit the number of requests made to an item on a network. Rate limiting helps to fight against denial of service (DoS) attacks by putting a cap on the number of pings that will be accepted by a resource (network, computer, server, etc) within a given time [9]. Often, what happens is if a device hits this cap in pings/requests, then the pings/requests will be dropped [9]. Regarding the ShieldsUp analysis that was performed, this is vital to understand. ShieldsUp sends out multiple pings to the first 1056 ports on a network device within a very short time. In doing this, depending on a network routers' configurations, it will likely drop the ping as a part of the rate limiting defence mechanism before the device can respond. When the ping is dropped, Shields Up registers this as a port in "stealth" mode rather than the actual configuration it is likely in, which is "closed". I believe that this is part a large part of the reason that each time I performed the service port scan, different ports showed as being closed compared to in stealth.

Ultimately, completing this project allowed me to revitalize knowledge and insight I had gained from the first time I ran the ShieldsUp port scans, while also acquiring new understandings of the topics being discussed. Completing this project a second time brought forth new questions that I did not have the first time, which took me down a very different research path than I had in the first assignment. I built upon the concepts of port

configurations that were discussed in CIDM6340 and expanded on how a single port can appear to be in two different states at once even though that is not possible. Furthermore, I was able to expand my knowledge on network and security defence mechanisms, which helped me to transform my thought process from being just about the results and what I can do to fix them, but to how these security measures work. Overall, there were many new insights I gained from completing this assignment a second time.

Resources:

- [1] **Hunt, T. (n.d.).** *Have I Been Pwned?* Retrieved October 11, 2025, from <https://haveibeenpwned.com/>
- [2] **Gibson Research Corporation. (n.d.).** *ShieldsUP!* Retrieved October 11, 2025, from <https://www.grc.com/x/ne.dll?rh1dkyd2>
- [3] **Fortinet. (n.d.).** *Firewalls.* Fortinet. Retrieved October 11, 2025, from <https://www.fortinet.com/resources/cyberglossary/firewall>
- [4] **Google. (2025).** Gemini (Version 2.5 Flash) [Large language model]. Response to "can a port be in two states at once?" prompt, October 4, 2025. <https://gemini.google.com/>
- [5] **Google. (2025).** Gemini (Version 2.5 Flash) [Large language model]. Response to "I have used ShieldsUp to run a port scanner twice on my laptop in two different periods of time over a year apart. the results were the same except for port 21, which was closed the first time I ran ShieldsUp but is currently showing stealth. Why is this?" prompt, October 4, 2025. <https://gemini.google.com/>
- [6] **Aldakkour, J. (2024).** *Shields Up and Nessus* [Unpublished manuscript]. College of Business, West Texas A&M University.
- [7] **Google. (2025).** Gemini (Version 2.5 Flash) [Large language model]. Response to "how does hibp work?" prompt, October 11, 2025. <https://gemini.google.com/>
- [8] **Google. (2025).** Gemini (Version 2.5 Flash) [Large language model]. Response to "how

come i get different results when I run shieldsup?" prompt, October 11, 2025.
<https://gemini.google.com/>

[9] Solo. (n.d.). *Rate limiting*. Solo. Retrieved from <https://www.solo.io/topics/rate-limiting>